Same Eye, Different Scanner: A Decision-Reliability Benchmark for Frontier VLMs and OCT Foundation Models in Glaucoma

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Abstract

A glaucoma-OCT reader used across providers should give the same eye the same diagnosis whatever scanner acquired the image. In practice, however, OCT measurements are not interchangeable across scanners, so a model's decision can change even though the eye is unchanged. We ask whether prompted frontier vision-language models (VLMs) and specialist OCT foundation models (FMs) keep a faithful, calibrated decision when the scanner changes. We build a benchmark that applies the scanner change as a same-eye image transform calibrated to measured cross-scanner disagreement. Because each eye is scored before and after, the scanner effect is de-confounded from patient differences while the images and labels stay real. We score the decision at a fixed operating point (a single decision threshold) by its decision-flip rate, change in sensitivity, and expected calibration error (ECE), with bootstrap confidence intervals (CIs) on the decision metrics. Four results follow. (1) Prompted frontier VLMs are the weakest and most overconfident above-chance readers on the raw B-scan (AUC 0.61–0.69). Their ECE is roughly 5–10$\times$ the best FM's for the Gemini and GPT families (Claude Opus 5 a better-calibrated exception); smaller open-weights and medical-specialist VLMs are near chance (0.41–0.59). (2) Rendered as a color thickness map, the stronger VLMs reach the level of the strong generic-FM and CNN readers (AUC 0.83–0.84), far above their raw-B-scan scores. The small open models do not recover, so what lets a VLM read the scan is the display format together with a capable base, not model size alone (a cross-dataset comparison). (3) A scanner change can break the decision while discrimination is intact: ΔAUC$\approx$0 while the diagnostic decision flips and specificity can collapse, invisible to AUC-only evaluation. (4) Reliability tracks the input representation and pretraining breadth, not model size: across a 26M–197M CNN sweep the largest model is steadiest on neither axis, and on the thickness-map axis a 98k-parameter from-scratch CNN matches 300M-parameter models on decision stability.

1 Introduction

Glaucoma is a leading cause of irreversible blindness, projected to affect 111.8 million people by 2040 [39] and often undiagnosed [38]. Optical coherence tomography (OCT) of the retinal nerve fiber layer (RNFL) is a standard structural test, recommended for diagnosis by clinical practice guidelines [12], and deep-learning readers now detect glaucoma from OCT at the level of specialists [34], increasingly via foundation models [48], producing one decision per eye: glaucoma or

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not. Prompted frontier vision-language models (VLMs) are being proposed as general-purpose medical readers [7, 33], attractive because they do not need task-specific training, but whether they are reliable enough for real deployment is largely untested.

That reliability is tested first by the scanner. Single-site care sidesteps the problem by repetition: the patient is monitored on one machine and simply re-scanned when referred elsewhere. To meet rising demand and prevent the follow-up delays that cause avoidable sight loss, clinical guidance is shifting glaucoma care to virtual, cross-provider triage in which a single digital OCT, acquired in community optometry [19], is shared and read remotely, with AI-assisted coordination [13]. Automated readers that support the decision are increasingly proposed to scale such screening and monitoring across providers [42], including teleophthalmology [40] and multi-site follow-up [30, 12]. OCT, however, is scanner-dependent: the same eye yields different RNFL measurements across scanners because segmentation and normative databases differ [6], and its raw B-scan appearance varies with frame-averaging, speckle, and contrast [36]. A reader deployed across providers must therefore treat the scanner as a nuisance variable that does not move the decision: one whose diagnosis tracks the scanner rather than the eye cannot support multi-provider care, whatever its single-scanner accuracy.

Single-scanner accuracy cannot see this, and existing work covers only parts of the problem. Cross-scanner claims for OCT foundation models are made on real cohorts where the scanner coincides with the patient group and are scored by accuracy or AUC alone [48, 25, 29]. Verbalized-overconfidence studies use natural images and no clinical decision [14], and medical-VLM trustworthiness benchmarks span modalities but omit any glaucoma decision under a scanner shift [46]. The closest glaucoma study fine-tunes VLM encoders on a single scanner, with no shift, no decision metric, and no calibration [18]. No prior benchmark combines a calibrated same-eye scanner shift, decision-level and calibration metrics, and a VLM-versus-FM common ground.

We therefore introduce a benchmark that measures this cross-scanner decision reliability directly. We apply the scanner change as a same-eye image transform calibrated to measured cross-scanner disagreement: published Bland–Altman limits of agreement for thickness maps [45, 4] and real per-vendor speckle for B-scans [36, 5]. Because the same eye is scored before and after, the scanner effect is de-confounded from patient differences, and the images and labels stay real. We score every reader at a fixed operating point by decision-flip, change in sensitivity, and calibration error, with bootstrap confidence intervals (CIs) on the decision metrics. On this common ground we place prompted frontier VLMs, smaller open-weights and medical-specialist VLMs, OCT and generic foundation models (FMs), a controlled ImageNet CNN model-size sweep, and a from-scratch CNN.

Contributions.

- **A de-confounded benchmark and decision-level metric suite for scanner-change reliability** (§3). To our knowledge the first evaluation that asks whether a glaucoma-OCT reader keeps its decision across scanners with the scanner effect isolated from patient differences, scored by a calibration-aware metric suite (decision-flip, Δsens@spec, spec@shift, ECE; bootstrap CIs on the decision metrics) that single-scanner accuracy cannot provide, applied uniformly across the VLM, foundation-model, and CNN readers.
- **Prompted frontier VLMs are the weakest and most overconfident above-chance readers of raw OCT** (§4, F1). On the raw B-scan they fall below the specialist FMs and are badly miscalibrated (with Claude Opus 5 a better-calibrated exception), and smaller open-weights and medical-specialist VLMs are near chance, an unsafe default for clinical OCT reading.
- **Frontier VLMs read a thickness map far better than a raw B-scan** (F2). The stronger frontier VLMs reach the strong generic-FM/CNN readers' level on the thickness-map task but fail on the B-scan task, pointing to input representation over model size or vendor; the gap is specific to general-purpose VLMs (a cross-dataset comparison we caveat in §5).
- **A scanner change can break the decision while discrimination holds** (F3). Discrimination can be unchanged (ΔAUC$\approx$0) while the decision flips and specificity can collapse, a failure AUC-only evaluation cannot detect and only decision-level and calibration metrics reveal.
- **Stability tracks representation and pretraining breadth, not model size** (F4). Across a one-family CNN model-size sweep, OCT and generic FMs, and a from-scratch network, tolerance to the scanner change is flat across model size: on the thickness-map axis a 98k-parameter CNN is CI-tied with the best CNNs and matches 300M-parameter models on decision stability.

2 Related work

Three lines of prior work are relevant to our question: the reliability of (medical) VLMs, cross-scanner generalization of OCT models, and prompted VLMs for glaucoma.

Reliability and evaluation of (medical) VLMs. Object-hallucination work showed that generative vision models assert ungrounded content [35, 23]. Verbalized-probability elicitation, the standard cross-model confidence denominator [24, 41, 47], is weakly calibrated for frontier multimodal models: Groot and Valdenegro-Toro [14] report GPT-4V and Gemini overconfidence on natural images, in line with the broader result that predictive uncertainty degrades under dataset shift [32]. On medical images, general-purpose and medical VLMs (e.g. LLaVA-Med [22]) remain well short of clinical-grade accuracy on broad medical benchmarks [7], reason weakly about eye images [33], and on glaucoma reach high accuracy but poor sensitivity on fundus grading [1]. Trustworthiness benchmarks such as CARES score an over-confidence ratio across medical modalities [46]. None of these pair a clinical decision with a calibrated scanner shift and a decision-level, calibration-aware metric, which is our focus.

OCT foundation models and cross-scanner generalization. OCT-specific foundation models claim cross-scanner transfer, but on real held-out cohorts where scanner coincides with cohort (a confound) and scored by accuracy or AUC [48, 25, 29]. A prior mitigation, cross-domain image translation, can recover device-specific glaucoma-OCT accuracy [3], but is validated by accuracy on real cohorts, not by decision stability under a same-eye shift. More broadly, distribution shift, both naturally occurring [20] and synthetic [16], is a mature benchmark topic, and retinal AI has been externally validated across fundus cameras [43], the same varying-hardware design we adopt. We differ by applying the scanner as a calibrated same-eye transform, which de-confounds it from cohort by construction, and by scoring the decision and its calibration rather than accuracy alone.

Prompted VLMs for glaucoma (the closest neighbor). Jalili et al. [18] fine-tune VLM vision encoders on a single scanner and find them competitive with OCT FMs (LLaVA 0.92 vs. RETFound 0.91), with no distribution shift, decision metric, or calibration. The same group's GPT-4V study [17] is the closest prompted-frontier-VLM precedent, but it grades fundus images and reports accuracy only, with no scanner shift or calibration. Our claim is instead about *prompted* frontier VLMs (zero- or few-shot, verbalized probability) under a calibrated scanner shift: a fine-tuned encoder can match a specialist while a prompted generalist stays weak and overconfident, so the two settings measure different deployment modes.

3 The benchmark

Common-ground harness. Every reader (a map-reader, an FM linear probe (LP) or fine-tune, or a prompted VLM) produces the same output: a per-eye glaucoma probability. We score each on paired clean→shifted degradation over the same test eyes under the same shared-seed shift, with eye-level percentile-bootstrap 95% CIs (2000 resamples; the eye is the unit of analysis; seed-0 split). Each result is labeled by its input representation, and we never cross-rank across axes.

Two calibrated axes (Fig. 1; validity in App. E). The two axes span the three most common OCT vendors (Cirrus, Spectralis, Topcon), which are known not to be interchangeable [6]. Each source vendor is fixed by its dataset; we take as the target the pair with a published calibration anchor (Cirrus→Spectralis for the thickness map, Spectralis→Topcon for the B-scan; App. E). The two axes therefore span two vendor pairs and two shift mechanisms, and the failure modes recur across both. The two axes are also the two input representations: the color thickness-map and the raw B-scan, so F2's representation contrast and the two shift axes are the same two settings. *Thickness-map axis*: Harvard-GDP Cirrus RNFLT maps [26] with labels (n=300 test, 46% glaucoma). The shift has three calibrated components: a per-eye bias and limit of agreement ($+1.94\,\mu$m, $\pm 7.68\,\mu$m, Warner-calibrated [45, 4]); a superior/inferior-wider sector profile from scan-circle centration [8]; and a spatially-correlated texture from OCT segmentation differences [28]. All are matched to real cross-vendor data (§E). *B-scan axis*: GLEAM *circumpapillary* Spectralis B-scans [44] with clinical labels (n=360 test, 49% glaucoma). The shift is a Spectralis→Topcon appearance stress: a frame-averaging change, injected multiplicative-Gamma speckle, and contrast dimming [36]. Its direction

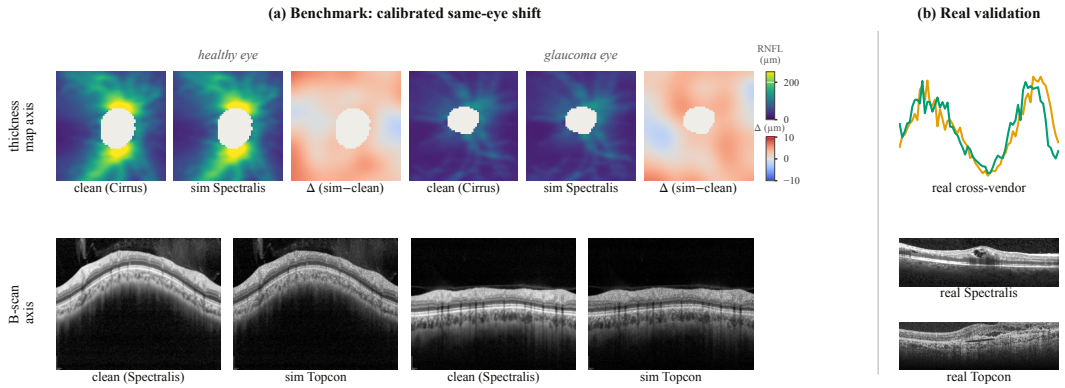


Figure 1: **The benchmark and its real-data validation.** **(a)** The calibrated same-eye shift on best-by-eye exemplars. One clear healthy and one clear glaucomatous eye are each shown clean → shifted on both axes: thickness maps (Harvard-GDP Cirrus, Warner-calibrated Cirrus→Spectralis; viridis, brighter = thicker RNFL) and circumpapillary B-scans (GLEAM Spectralis → simulated Topcon). The shift is small by design, so each clean/shifted map pair is followed by a difference panel (sim − clean, coolwarm, on a shared $\pm 10\,\mu$m scale spanning the calibrated LoA), with the largest deviations in the superior and inferior nerve-fiber bundles, where the calibrated cross-scanner shift is widest (§E). This is one representative per-eye realization; the benchmark applies an independent calibrated shift to every eye. Each reader sees clean and shifted versions of the same eyes, so scanner is not confounded with cohort. **(b)** Real data corroborating the shift: cross-vendor cpRNFL (circumpapillary RNFL) profiles from GLEAM Spectralis and GOALS Topcon agree in shape, and RETOUCH Spectralis/Topcon macular B-scans calibrate the per-vendor speckle of the B-scan axis.

and magnitude are calibrated from real per-vendor RETOUCH macular display speckle [5]. This axis tests tolerance to appearance changes: its magnitude is speckle-matched, not bounded by a measured cross-scanner LoA, which would require same-eye disc-OCT on two scanners that no public dataset provides.

Decision-level metrics. We report discrimination, decision, and calibration reads; clean AUC, decision-flip, and Δsens@spec carry eye-level 95% bootstrap CIs, and the remaining reads are point estimates. *Discrimination:* clean AUC and ΔAUC (signed clean−shift). *Decision:* decision-*flip* [16], the fraction of eyes whose decision (refer / do-not-refer) changes under the shift; Δsens@spec, the clean−shift change in sensitivity at the operating point; and *spec@shift*, the specificity actually realized under the shift, which flags a reader that keeps its flip low only by collapsing specificity. *Calibration:* ECE and ΔECE [15] over 10 equal-width bins, and the net calibration error (NCE), a signed calibration error [14]: mean predicted-class confidence minus accuracy, so $+$ = overconfident. The operating point is a single threshold fixing 90% specificity on clean negatives, applied to shifted scores; fixing it on the clean data makes clean specificity slightly optimistic, but equally for every reader, so comparisons are unaffected. Because a recalibrating shift leaves ΔAUC$\approx$0 while the decision flips, it is flip, Δsens, and spec@shift, not ΔAUC, that reveal the failure.

Readers. *Frontier closed VLMs* (three developers): Gemini (2.5 Flash, 2.5 Pro, 3.6 Flash), GPT-5.6 Sol, and Claude Opus 5, zero- and few-shot (k=6, single greedy sample; temperature-0 where the API accepts it, App. A), on both the raw B-scan and a color thickness-map render. *Open-weights and specialist VLMs* (self-hosted): Qwen3-VL-8B [2] and MedGemma-1.5-4B [37], same protocol. Elicitation is verbalized probability [41]: each model states its probability of glaucoma in words rather than reading it from output logits, the one score common to every vendor, since the closed models expose no logprobs. *OCT and generic FMs:* RETFound [48] and MIRAGE [29] against DINOv2 [31], LP and fine-tune. *ImageNet CNN model-size sweep:* ResNet-50 and ConvNeXt-T/B/L (26M→197M), frozen probe and per-architecture fine-tune. *From-scratch:* a 98k-parameter SmallCNN trained end-to-end on each axis.

Table 1: **Headline configurations, both axes** (calibrated shift, eye-level bootstrap 95% CIs). Columns: clean AUC, decision-*flip*, Δsens@spec (signed clean$-$shift), clean ECE, net calibration error (NCE) and ΔAUC. Lower flip, $|\Delta\text{sens}|$, ECE and $|\Delta\text{AUC}|$ are better. **Bold** is the best AUC, flip and ECE per column among readers with useful discrimination (clean AUC ≥ 0.70); a near-chance reader can post a trivially low flip, so flip is read together with AUC. NCE is a signed calibration error [14] (mean predicted-class confidence minus accuracy; $+ =$ overconfident); Δsens, NCE and ΔAUC are signed diagnostics ($\approx$0 ideal) and are not bolded. The ImageNet-CNN row is the best per axis; the from-scratch SmallCNN is trained end-to-end on each axis and never sees the test shift. VLMs are prompted (*fs* = few-shot). Full per-axis tables (B-scan 29 configurations, map 25), with realized spec@shift and flip/Δsens CIs, are in App. H.

Reader	Class	AUC [95%, CI]	Flip↓	Δsens	ECE↓	NCE	ΔAUC
B-scan axis	Spectralis $\rightarrow$ Topcon, n=360 eyes						
DINOv2 · FT	generic FM	**0.88** [0.84, 0.91]	**0.09**	+0.08	0.05	−0.00	+0.01
RETFound · FT	OCT FM	0.85 [0.80, 0.89]	0.14	−0.09	0.07	+0.07	−0.01
MIRAGE · LP	OCT FM	0.70 [0.65, 0.76]	0.47	−0.72	**0.04**	−0.02	+0.04
ConvNeXt-B · LP	ImageNet CNN	0.83 [0.78, 0.87]	0.11	+0.07	0.07	+0.03	+0.04
SmallCNN (98k)	scratch CNN	0.79 [0.74, 0.83]	0.27	−0.32	**0.04**	−0.04	+0.02
Gemini 3.6 Flash *fs*	frontier VLM	0.69 [0.64, 0.74]	0.15	+0.01	0.26	+0.26	−0.00
GPT-5.6 Sol *fs*	frontier VLM	0.67 [0.61, 0.73]	0.24	+0.11	0.23	+0.23	−0.01
Claude Opus 5 *fs*	frontier VLM	0.67 [0.62, 0.73]	0.18	+0.05	0.16	+0.13	+0.03
Qwen3-VL-8B *fs*	open VLM	0.54 [0.49, 0.58]	0.02	+0.00	0.38	+0.38	+0.00
MedGemma-1.5-4B *fs*	specialist VLM	0.49 [0.43, 0.54]	0.00	+0.03	0.42	+0.42	−0.04
Thickness-map axis	Cirrus $\rightarrow$ Spectralis, n=300 eyes						
DINOv2 · FT	generic FM	**0.88** [0.83, 0.91]	0.04	+0.01	0.09	−0.02	+0.00
ConvNeXt-L · FT	ImageNet CNN	0.87 [0.83, 0.91]	**0.02**	+0.01	**0.07**	−0.04	+0.00
SmallCNN (98k)	scratch CNN	0.85 [0.80, 0.89]	**0.02**	+0.00	0.12	+0.03	−0.01
GPT-5.6 Sol *fs*	frontier VLM	0.84 [0.79, 0.88]	0.09	+0.02	0.11	+0.11	−0.00
Gemini 3.6 Flash *fs*	frontier VLM	0.84 [0.79, 0.88]	0.05	+0.06	0.15	+0.13	+0.02
Claude Opus 5 *fs*	frontier VLM	0.83 [0.78, 0.88]	0.09	+0.03	**0.07**	+0.01	−0.00
Qwen3-VL-8B *fs*	open VLM	0.59 [0.55, 0.62]	0.05	+0.00	0.36	+0.36	+0.02
MedGemma-1.5-4B *fs*	specialist VLM	0.55 [0.48, 0.61]	0.03	+0.00	0.30	+0.30	+0.02

4 Experiments and findings

We score 29 configurations on the B-scan axis and 25 on the thickness-map axis (the OCT-specific B-scan FMs are not run on map renders; setup and model/dataset cards in App. A–C). Each finding below states one claim and gives its eye-level bootstrap-CI evidence; cross-configuration comparisons are descriptive and uncorrected. Table 1 lists the headline configurations. Figs. 2–3 give the reliability landscape and the shift mechanism.

F1: prompted frontier VLMs are the weakest and most overconfident above-chance readers, across three developers. Among the readers above near-chance, the prompted frontier VLMs rank last. On the raw B-scan, even the best frontier VLM only matches the weakest specialist's discrimination (MIRAGE · LP, AUC 0.70): Gemini 3.6 Flash few-shot AUC 0.69 [0.64, 0.74], GPT-5.6 Sol 0.67 [0.61, 0.73], and Claude Opus 5 0.61/0.67 (zero/few-shot). The Gemini and GPT configurations are also badly overconfident: their clean (pre-shift) ECE is 0.23–0.42, roughly 5–10$\times$ the best FM's 0.04, and their net calibration error is also large and positive (0.23–0.42; per configuration in Table 1 and App. H), so the miscalibration is overconfidence, not two-sided noise. The gap is not a binning artifact: it holds under a strictly proper score (Brier) and adaptive-bin ECE, not just fixed-bin ECE (App. H), matches Groot and Valdenegro-Toro [14], Xia et al. [46], and is the VLM/FM split in Fig. 2c. Few-shot lifts discrimination with mixed calibration effect, whereas neither a larger model nor a higher tier helps (the higher-tier Gemini 2.5 Pro is the worst frontier zero-shot, AUC 0.51). Calibration is vendor-dependent, with Claude Opus 5 a markedly better exception (ECE 0.11–0.16). The weakness deepens as models shrink: the open-weights Qwen3-VL-8B and the specialist MedGemma-1.5-4B are near chance on both axes (AUC 0.41–0.59) with the highest ECE (up to 0.51), so neither open weights nor medical-specialist pretraining rescues the decision (the small-model elicitation caveat in §5 applies).

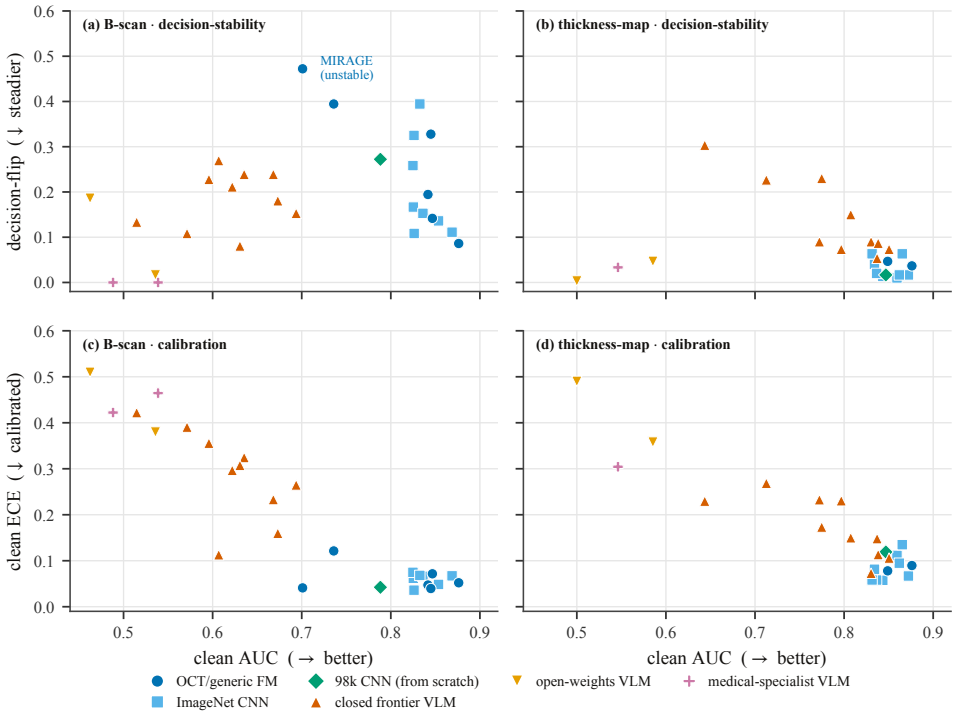


Figure 2: **Reliability landscape.** Every reader on clean AUC against two reliability metrics, *decision-stability* (decision-flip, top row) and *calibration* (clean ECE, bottom row), for the B-scan (left) and thickness-map (right) axes. Color and marker encode reader class. Frontier VLMs are unstable and overconfident on the raw B-scan (a,c) but recover on the thickness map (b,d); the small open and specialist VLMs are near chance on both axes. A ΔAUC$\approx$0 can hide large decision-flips (MIRAGE, panel a), and the 98k-parameter from-scratch CNN (green diamond) is among the steadiest readers on the thickness-map axis (b), CI-tied with the best CNNs, though not steady on the raw B-scan (a).

F2: general-purpose VLMs read the thickness map, not the raw B-scan. On the thickness-map task the same models reach the strong generic-FM/CNN readers' level (DINOv2, ConvNeXt): Gemini 3.6 Flash 0.84, GPT-5.6 Sol 0.84 [0.79, 0.88], and Claude Opus 5 0.83 [0.78, 0.88]. Discrimination, calibration (ECE roughly halved), and stability (flip dropping) recover together, well above their raw-B-scan level and across all three developers (per-VLM detail in App. F). The lift holds even zero-shot (GPT-5.6 Sol 0.77 vs. 0.63; Claude Opus 5 0.61$\rightarrow$0.85), so it is not few-shot memorization, but it needs a capable base: the small open VLMs do not recover (Qwen3-VL-8B 0.50–0.59, MedGemma-1.5-4B 0.41–0.55). The gap is specific to general-purpose VLMs on the raw B-scan, where reading RNFL requires the implicit segmentation an OCT-trained model performs while a color thickness map presents thinning directly. A thickness map is not intrinsically a better input (trained models read glaucoma from raw OCT at least as well [27]), and because the two axes use different datasets this comparison is confounded (§5).

F3: a scanner change can break the decision while discrimination holds. Decision-flip is unrelated to discrimination: across all 29 B-scan readers $|\Delta\text{AUC}| \leq 0.08$ (about half within 0.03), yet decision-flip spans 0.00–0.47 and shifted specificity 0.07–1.00, and flip is uncorrelated with ΔAUC (Spearman 0.09). Flip is instead a threshold effect: it rises with the mean absolute per-eye movement mean $|\overline{\Delta p}|$ (Spearman 0.88; movement $\Delta p = p_{\text{shift}} - p_{\text{clean}}$) more tightly than with the net directional slide $|\overline{\Delta p}|$ (0.68; Fig. 3a,b), because some readers move their scores two-sidedly and flip without a net direction. What matters for safety is the direction of the crossings. In the appearance-sensitive readers (the OCT FMs, the fine-tuned ImageNet CNNs, and the from-scratch SmallCNN) the added speckle pushes scores up, so essentially every threshold crossing is toward "refer" and specificity collapses; the up-shift largely preserves the ordering of eyes, so AUC holds. MIRAGE · LP is the extreme (all 170 flips up, specificity 0.07), with RETFound · LP (117/118 up) and ResNet-50 · FT

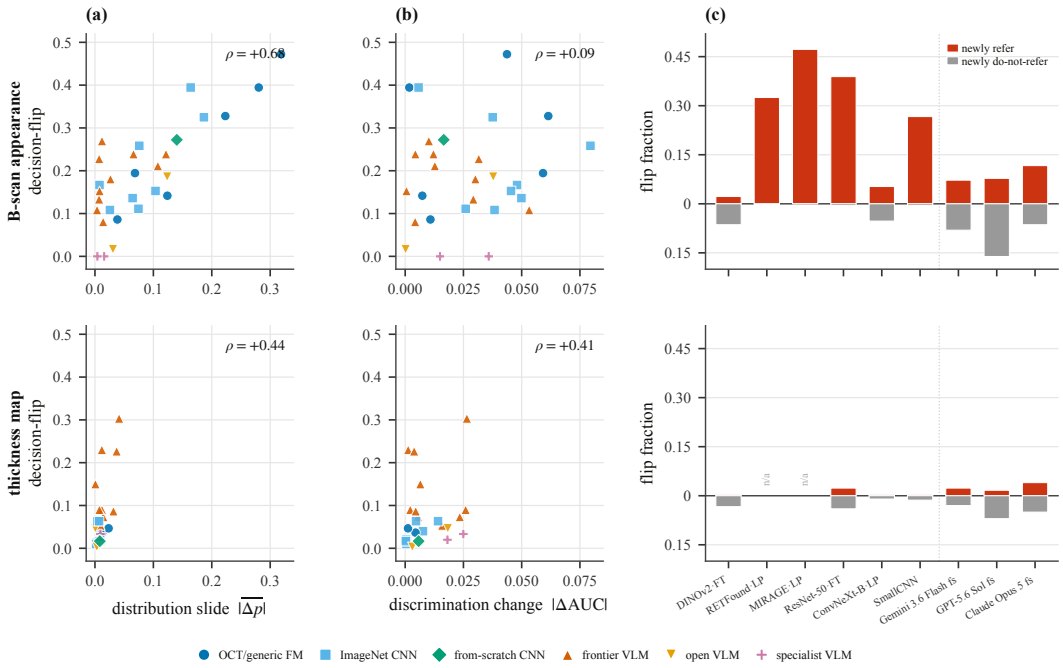


Figure 3: **Decision-flip comes from the score distribution moving, not from a change in discrimination.** One row per shift axis (top: B-scan appearance; bottom: thickness map), with matched limits. Across all 29 B-scan readers, decision-flip is uncorrelated with the discrimination change $|\Delta\text{AUC}|$ (**b**, ρ=0.09) and rises with the net directional slide $|\overline{\Delta p}|$ (**a**, ρ=0.68); readers well above the (**a**) trend (real flip at near-zero net slide) move two-sidedly, and flip tracks the mean absolute per-eye movement mean $|\Delta p|$ (not plotted) even more tightly (ρ=0.88). A reader can flip many decisions with its AUC intact. On the thickness-map axis (bottom) slide, $|\Delta\text{AUC}|$, and flip are all small. **(c)** Decision-flip split into newly-"refer" (over-referral, red) and newly-"do-not-refer" (grey), for representative readers per class plus the frontier VLMs (few-shot; the dotted line separates non-VLM from VLM). The appearance-sensitive readers (the OCT FMs MIRAGE · LP and RETFound · LP, the ImageNet ResNet-50 · FT, and the from-scratch SmallCNN) over-refer almost one-directionally, whereas the generic DINOv2 · FT and ConvNeXt-B · LP stay robust (small, two-sided). The VLMs flip in *model-specific* directions (GPT-5.6 Sol toward do-not-refer, Claude Opus 5 toward refer, Gemini 3.6 Flash either way). The map row (bottom) is likewise near-flat for every reader, and the OCT-specific FMs, not run on maps, are marked n/a. Markers in (a,b) encode reader class (shared legend).

(140/142 up) the same, and 18/29 readers net-shift toward refer (OCT FMs mean $\overline{\Delta p} = +0.24$). This is over-referral, not lost sensitivity (Fig. 3c), and it concentrates on healthy eyes (MIRAGE · LP flips 62% of healthy vs. 28% of moderate-to-advanced; no disparity across demographics, underpowered subgroups, App. G). The frontier VLMs and the broadly-pretrained DINOv2 show no such systematic over-referral: the VLMs flip in model-specific directions (GPT-5.6 Sol toward do-not-refer, Claude Opus 5 toward refer, Gemini 3.6 Flash either way; flip 0.15–0.24), and DINOv2 resists ($\overline{\Delta p} = -0.04$), the steadiest capable reader (flip 0.09). Either way AUC-only evaluation misses the flip while the decision-level metrics expose it. The thickness-map axis is far more stable ($|\Delta\text{AUC}| \leq 0.03$, flip ≤ 0.31).

F4: stability does not track model size; representation and pretraining breadth win. Frozen LP clean AUC is unchanged across 26M→197M (map 0.83–0.84, B-scan 0.82–0.83; CIs overlap), and the largest model is steadiest on neither axis; the color thickness-map is robust at every size while the raw B-scan is not. Generic DINOv2 is the steadiest and best-discriminating (B-scan LP→FT flip 0.19→0.09), whereas OCT-specific MIRAGE is weakest (AUC 0.70, flip 0.47); fine-tuning rescues the other OCT FM (RETFound · FT flip 0.14, AUC 0.85), so the weakness is MIRAGE-specific, not intrinsic to OCT pretraining, though fine-tuning is no universal fix and can itself distort features

under shift [21]. The clearest case is a from-scratch 98k-parameter SmallCNN, trained end-to-end on the thickness map and never exposed to the test shift: it flips just 0.02 $[0.00, 0.03]$ (net calibration error 0.03), CI-tied with the best ImageNet CNNs and steadier than the 300M DINOv2 (flip 0.04) at $\sim$900–3000$\times$ fewer parameters, so its stability is a genuine out-of-distribution result from the compact, task-trained representation, not model size.

5 Discussion and limitations

What this means for deployment. Used in a glaucoma-diagnosis pipeline today, a prompted generalist VLM reading raw B-scans is the least safe above-chance reader: below the specialist FMs and markedly overconfident. But a strong FM or CNN fails differently: it holds its AUC under a scanner change while its decisions flip, so it passes single-scanner validation and then fails in the field. Where that flip is a specificity collapse, it over-refers the healthy majority of a screening population (§4), the costliest place for false referrals. Two things follow. First, this is a decision failure, not a calibration one: calibration is already poor before any shift and does not significantly worsen under it (ΔECE CI includes 0), and post-hoc temperature scaling [15] removes most of the clean VLM ECE (App. D) yet, being monotonic, leaves the decision-flip unchanged. Second, re-thresholding cannot undo this score movement at deployment, where no labeled data from the incoming scanner exist, so the remedy is to select readers whose decisions stay put. The recommendation is concrete: validate a diagnostic reader at its operating point, with calibration and under a scanner shift, not on single-scanner AUC; our benchmark is that check, and a candidate that flips a quarter of decisions under a calibrated same-eye shift is unsafe however high its AUC.

What to engineer, and why. The failure is not fundamental to VLMs. The same models recover to the strong generic-FM/CNN level once the RNFL is rendered as a readable color thickness map (a cross-dataset contrast; see *Limitations*). What helps is the input representation, not model size, vendor, or a "medical-specialist" label. The B-scan failure is an appearance sensitivity, consistent with the added speckle and dimmed contrast being read as RNFL thinning: appearance-keyed readers (the OCT-specific FMs and fine-tuned CNNs) raise their glaucoma score and over-refer while the broadly-pretrained generic FM resists (F3–F4); the small open and specialist models recover on neither, so it is an interaction of input format and capability. Robustness tracks representation and pretraining breadth, not model size: a 98k-parameter task-trained CNN is among the steadiest readers at $\sim$900–3000$\times$ fewer parameters than the FMs. For practitioners: engineer the input representation and a compact, task-specific encoder, and require decision stability, not model size, before deployment.

Limitations. *The same-eye shift is synthetic*: no public same-eye cross-scanner labeled disc-OCT exists, so we calibrate the shift to published cross-scanner LoA (map) and real per-vendor speckle (appearance) and keep four datasets real; a real same-eye cross-scanner set is the natural extension. *Elicitation and calibration are conservative*: VLMs are queried with a single greedy verbalized-probability sample, so the reported VLM ECE is an upper estimate of overconfidence (N-sample, prompt-paraphrase passes are future work); the smallest open models collapse to a constant (Qwen3-VL-8B on the map), so a boolean-verdict elicitation may be fairer. *The B-scan-vs-thickness-map comparison (F2) is cross-dataset*: the two axes differ in dataset, vendor, and labels, so F2 confounds input representation with dataset difficulty and possible contamination; we read it as reader-specific (§4), not as evidence a map is better in general. *Subgroups*: instability is broadly uniform across disease severity and demographics (App. G), but the subgroups are underpowered and image quality is unrecorded, so a definitive fairness audit needs richer metadata. *Scope*: two axes, never cross-ranked; broader vendor-pair coverage and extending decision stability to longitudinal progression monitoring are the natural next steps.

6 Conclusion

A scanner change can break a glaucoma-OCT reader's decision while its AUC looks unchanged; reliability tracks input representation and pretraining breadth, not model size. We release our deconfounded, decision-level, calibration-aware benchmark (four real datasets, only the scanner shift synthesized) as a reusable check, so a reader's decision stability under a scanner change can be validated before it reaches patients.

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A Experimental setup

Encoders. Foundation models (RETFound, MIRAGE, DINOv2) and CNNs (ResNet-50, ConvNeXt-T/B/L, the 98k-parameter SmallCNN) are frozen feature extractors with a linear head, or fine-tuned end-to-end ("FT"). *Linear probe:* `StandardScaler` + ℓ_2 logistic regression, `LogisticRegressionCV` ($C \in \{0.01, 0.05, 0.2, 1.0\}$, 3-fold, ROC-AUC, `max_iter`=5000; small-n fallback C=0.05), fit on clean training eyes (seed-0). Features: DINOv2 = `[CLS]` $\oplus$ mean patch tokens (registers excluded) at 224^2; RETFound = global-avg-pool + `fc_norm` at 224^2 (ImageNet normalization); MIRAGE = encoder last token at native 512^2 (unit-scaled, not ImageNet-normed); ResNet/ConvNeXt = global-avg-pool. *Fine-tune:* AdamW (weight decay 0.05), 50 epochs, batch 32, layer-wise LR decay (ConvNeXt 0.9 / ResNet 1.0 / ViT 0.65), base-LR 5×10^{-4}/10^{-3}/5×10^{-3} linearly scaled by batch, warmup then cosine to 10^{-6}, class-weighted cross-entropy with label-smoothing 0.1, `RandomResizedCrop`(0.5–1.0)+HFlip+RandomErasing(0.25) at 224^2, mixed precision, early stop (patience 12) on validation AUC. *SmallCNN* ($\sim$98k params): four `Conv`3$\times$3–GroupNorm–ReLU–MaxPool blocks (16/32/64/128) $\rightarrow$ global-avg-pool $\rightarrow$ linear logit; Adam (10^{-3}, wd 10^{-4}), class-weighted BCE. It is trained end-to-end from scratch on each scored axis (GLEAM B-scans; Harvard-GDP thickness maps), on the same seed-0 split as the other readers. It is never exposed to the test shift.

VLMs. All seven VLMs receive byte-identical eyes, shifts (per-eye seeds $3000{+}e$, $4000{+}e$), few-shot exemplars (seed $+11$) and prompts from a single shared axis builder, are queried greedily (one sample) for a verbalized probability [41], and are thresholded at the clean-data 90%-specificity operating point. Exact model IDs and inference settings are in Table 2. The verbatim prompts, few-shot construction and probability parsing are in App. B.

Statistics. Confidence intervals are eye-level percentile bootstraps (2000 resamples; the eye is the resampling unit) [9]. ECE uses 10 equal-width confidence bins [15]. The operating point (target specificity 0.90) is fixed on clean negatives and reused on shifted scores; Δsens@spec and spec@shift are read there, while decision-flip is the change in the fixed 0.5 clinical call (refer / do-not-refer). Both capture decision movement at a deployed threshold. Sample sizes: n=360 eyes (B-scan axis), n=300 eyes (thickness-map axis).

Compute. The open-weights VLMs and all fine-tuning run on a single Colab GPU (A100-40GB or L4), bf16, `device_map=auto`, batched greedy generation. The closed API configurations (Gemini, GPT-5.6 Sol, Claude) are CPU-only clients. Reproducing the tables and figures from the released per-eye outputs requires no GPU.

B Prompts and elicitation

Elicitation is a single verbalized referral probability per eye [41], the score common to every vendor, since the closed models expose no logprobs. Prompts are byte-identical across models on a given axis.

B-scan axis. System: *"You are grading circumpapillary OCT B-scans for glaucoma. Each image is a cross-sectional retinal scan taken in a ring around the optic nerve head. Glaucoma shows as thinning/loss of the retinal nerve fibre layer (RNFL) and neuroretinal rim."* Query: *"Now assess THIS new B-scan. Return JSON:* `glaucoma_referral_probability` *(0–1, your calibrated probability that this eye should be referred for glaucoma),* `rationale` *(one short sentence)."*

Thickness-map axis. System: *"You are grading en-face retinal nerve fibre layer (RNFL) THICKNESS MAPS for glaucoma. Each image is a colour-coded map of RNFL thickness around the optic disc: warm colours = THICK, cool colours = THIN. Glaucoma shows as THINNING—loss of the bright superior and inferior nerve-fibre bundles (the 'double-hump'), a wedge defect or diffuse darkening."* Query: as above with "RNFL thickness map".

Few-shot. k=6 labeled exemplars, drawn only from the training pool (leakage guard), balanced: B-scan axis severity-graded (3 healthy + one each early/moderate/advanced); map axis binary (3 glaucoma + 3 healthy). Selection RNG is `seed`$+11$ and the exemplar set is identical across models.

Table 2: **Model card.** Exact identifiers, size and inference settings. VLMs are queried $N{=}1$, greedily. Reasoning/thinking is enabled where supported ("high"/dynamic). Temperature is omitted for the reasoning API models (they reject it) and set to 0 for Gemini.

Model	Identifier	Params / settings
Vision-language models (prompted)		
Gemini 2.5 Flash / Pro	`gemini-2.5-flash`, `-pro`	temp 0, thinking budget
Gemini 3.6 Flash	`gemini-3.6-flash`	temp 0, thinking level high
GPT-5.6 Sol	`gpt-5.6-sol`	reasoning=high, no temp
Claude Opus 5	`anthropic/claude-opus-5`	reasoning=high
Qwen3-VL-8B	`Qwen/Qwen3-VL-8B-Instruct`	8B, bf16, greedy, 192 tok
MedGemma-1.5-4B	`google/medgemma-1.5-4b-it`	4B, bf16, greedy, 192 tok
Foundation models / CNNs (frozen probe + fine-tune)		
DINOv2 (ViT-L/14, reg.)	`vit_large_patch14_reg4_dinov2`	$\sim$300M, 224^2
RETFound (ViT-L/16)	`RETFound_mae_natureOCT`	$\sim$300M, 224^2
MIRAGE-Base (ViT-B)	`j-morano/MIRAGE-Base`	$\sim$86M, 512^2
ResNet-50	`resnet50.a1_in1k`	$\sim$26M
ConvNeXt-T/B/L	`convnext_{t,b,l}.fb_in1k`	$\sim$29/88/197M
SmallCNN	from scratch	$\sim$98k

Table 3: **Datasets.** Two are scored (thickness-map and B-scan axes). Two supply the real-data validation (cross-vendor anatomy; per-vendor speckle calibration).

Dataset	Role	Modality (region)	n (tr/val/test)	Label / access
Harvard-GDP	thickness-map axis (scored)	Cirrus RNFLT maps 225^2 (disc)	550/150/300	VF-derived; CC BY-NC-ND 4.0
GLEAM	B-scan axis (scored)	Spectralis circumpapillary B-scans	660/180/360	clinical (4-class$\rightarrow$binary)
GLEAM / GOALS	cross-vendor validity	Spectralis + Topcon B-scans $\rightarrow$ cpRNFL	60+60	single segmenter [10]
RETOUCH	speckle calibration	Cirrus/Spec/Topcon macular B-scans	$\sim$30/vendor	public challenge

The few-shot preamble adds *"Below are LABELLED reference images in this exact format so you can calibrate"*; each exemplar is captioned with its label before (API/open configurations) or after (Gemini) its image.

Probability parsing. We read `glaucoma_referral_probability` from the JSON (regex fallback on the first such field, clipped to $[0, 1]$); parse failures are retried (up to 5, exponential backoff) and, only if still failing, imputed as 0.5 (never cached). Imputation is rare for the frontier configurations but non-negligible for the small-model configurations that collapse to a near-constant output (e.g. Qwen3-VL-8B on the map), where it biases ECE and AUC toward chance and reinforces the near-chance reading of those configurations. An alternative boolean-verdict schema maps $p{=}$conf if glaucoma else $1{-}$conf. We use the probability schema throughout for parity.

C Datasets

All data are public or registered-access research datasets. No new patient data were collected. Splits are eye-level (one eye per subject where applicable), seed-0, 55/15/30 train/val/test.

Harvard-GDP labels are the `glaucoma` field per eye [26]; GLEAM is single-center [44], a 4-class severity label binarized to refer/do-not-refer. The cross-vendor validity row is a *derived measurement*, not a released dataset: we run the device-agnostic ONH-OCT segmenter of Driessen et al. [10] on GLEAM Spectralis and GOALS Topcon [11] B-scans to produce cpRNFL profiles, holding the segmentation source constant so the residual cross-vendor difference is imaging plus cohort, not segmentation. RETOUCH [5] is used only to measure real per-vendor display-speckle contrast (Cirrus 0.365 $>$ Topcon 0.184 $>$ Spectralis 0.148), not scored. It is macular data, a proxy for the disc. No public same-eye cross-scanner labeled disc-OCT set exists, which is why the primary shift is a calibrated same-eye transform (§E).

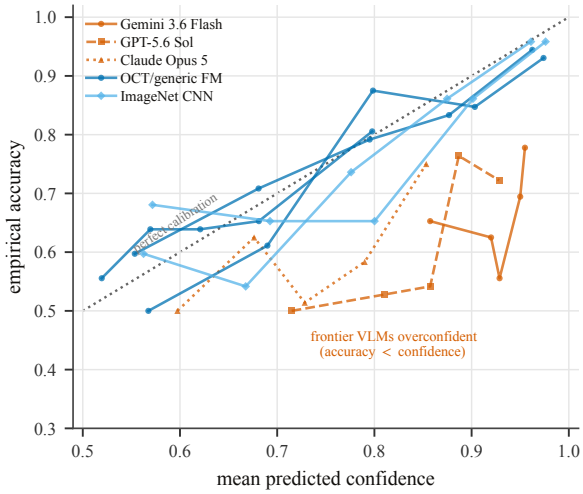


Figure 4: **Reliability diagram** (B-scan axis, few-shot), one line per configuration: empirical accuracy vs. mean predicted confidence in five equal-count bins for readability (dotted = perfect calibration; the ECE metric itself uses 10 equal-width bins, §A). Every frontier VLM (Gemini 3.6 Flash, GPT-5.6 Sol, Claude Opus 5; one warm color, distinguished by line style) lies below the diagonal (overconfident), most for Gemini and least for Claude, whereas the OCT/generic FMs and ImageNet CNNs (cool colors) track it (well-calibrated).

D Reliability diagrams (calibration detail)

Fig. 4 shows the ECE gap of §4 (F1) is genuine overconfidence, not a binning artifact, and systematic across vendors: every frontier-VLM configuration (Gemini, GPT-5.6 Sol, Claude Opus 5, few-shot) lies below the diagonal (confidence exceeding accuracy) while the OCT/generic FMs and ImageNet CNNs track it. Vendor-dependence is visible (Gemini furthest below, Claude closest). The FM/CNN per-eye probabilities here are recomputed from cached encoder features with the same frozen probe (§A), and the recomputed ECE matches Table 1.

Net calibration error and post-hoc calibration. The miscalibration is directional and largely fixable post hoc, yet the decision failure is not. For each frontier configuration we compute the net calibration error (NCE), a signed calibration error [14]: mean decision-confidence ($\max(p, 1-p)$) minus decision accuracy on clean eyes (positive = overconfident; the opposite sign to Groot et al.'s accuracy−confidence convention). (The ECE in Table 1 is instead the referral-probability, or positive-class, calibration error; the NCE and the reliability diagram use the predicted-class confidence $\max(p, 1-p)$. Both show the frontier VLMs are badly miscalibrated.) The Gemini and GPT configurations have NCE 0.23–0.42 (few-shot 0.23–0.32, zero-shot up to 0.42); Claude Opus 5 is far smaller (0.08–0.13), matching the diagram ordering above. Post-hoc temperature scaling [15] (one scalar fit on a held-out clean half by minimizing NLL) removes most of the clean ECE (on the held-out clean half: Gemini 3.6 Flash few-shot $0.27 \to 0.07$, GPT-5.6 Sol $0.26 \to 0.07$), but, being strictly monotone, it preserves the score ordering and fixes the 0.5 decision point, so the decision-flip is exactly unchanged (and the refit 90%-specificity operating point selects the same decisions). Calibration is a post-hoc fix; the scanner-driven decision-flip is not.

Calibration beyond fixed-bin ECE. Fixed-bin ECE is a biased estimator at $n \approx 300$–360 and depends on the binning scheme, so we recompute the clean-eye calibration under two complementary reads with different failure modes: adaptive (equal-count) ECE, which debiases the fixed binning, and the Brier score, a strictly proper scoring rule with no binning (the proper-score check used under dataset shift by Ovadia et al. [32]). Both are reported for every reader on both axes as extra columns in the full tables (App. H). The frontier VLMs stay worse than every FM/CNN reader on all three reads (fixed-bin ECE, adaptive ECE, and Brier; Claude Opus 5 the least-overconfident exception), so the overconfidence gap is not an artifact of the ECE binning.

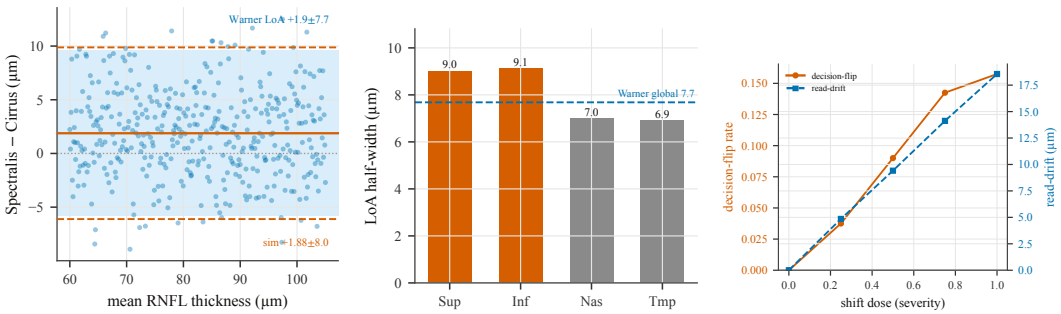


Figure 5: **Thickness-map shift is LoA-calibrated.** (a) Bland–Altman of the synthetic per-eye differences against Warner's published 95% limits (band); (b) limits widen in the superior/inferior sectors; (c) decision-flip and read-drift rise monotonically with shift dose, a graded axis rather than a cliff.

E Shift calibration and real-data corroboration

The synthetic shift is magnitude-tied to the published cross-scanner agreement literature and direction-corroborated by real data. *Thickness-map axis:* the Cirrus→Spectralis RNFL offset reproduces Warner et al. [45]'s $+1.94\,\mu$m bias and $\pm 7.68\,\mu$m 95% limits (difference SD $3.92\,\mu$m, Fig. 5a, a Bland–Altman plot [4]). The per-eye spread is anatomy-coupled, drawn in quadrature from segmentation, centration, and magnification mechanisms (between-eye SDs $2.8/2.3/1.5\,\mu$m) with superior/inferior-wider sector gains, so limits widen in the vulnerable sectors (Fig. 5b). The magnitude is calibrated to Warner's Bland–Altman statistics, and the superior/inferior-wider spatial profile follows the dominant cross-device error mechanism, scan-circle centration, which perturbs the superior and inferior sectors most [8]; the three mechanism components sum in quadrature to the published SD by construction, so both the shift magnitude and its sector direction are anchored in measured cross-scanner data and established optics. The finer within-eye texture is a small mean-zero term for spatially-correlated segmentation differences, which are largest in the superior and inferior sectors [28]. This smoothness is data-driven: the real cross-vendor cpRNFL difference (GLEAM Spectralis vs GOALS Topcon) is itself spatially smooth, with 74% of its angular variance in the lowest five harmonics and a $\sim 22^\circ$ correlation length (roughness 0.79 vs 1.41 for white noise), i.e. a low-frequency difference, not per-pixel noise. Being mean-zero, the texture leaves the calibrated between-eye bias and LoA intact; these are validated directly by the Bland–Altman fit (Fig. 5a) and the monotone dose–response (Fig. 5c), so Fig. 1's difference panels illustrate the calibrated shift rather than assert a per-pixel measurement. Decision-flip and read-drift rise monotonically with dose (Fig. 5c). *B-scan axis:* the Spectralis→Topcon stress changes frame-averaging (16→1 frames), injects a multiplicative-Gamma speckle field that lifts display-speckle contrast $0.148 \rightarrow 0.184$ (matched to real RETOUCH, Fig. 6a), adds near-inert axial blur ($\sqrt{15}\,\mu$m FWHM at $10\,\mu$m/px), and dims backscatter contrast $\times 0.773$. Real cross-vendor cpRNFL corroborates that disc anatomy is vendor-consistent: with a single segmenter [10], cpRNFL profiles from GLEAM Spectralis and GOALS Topcon [11] B-scans agree in shape ($r \approx 0.93$, Fig. 6b). This corroborates that the scanner perturbs the measurement, not the anatomy; it is a shape agreement, not a claim of scanner-interchangeability.

F Thickness-map vs. raw-B-scan reading, per VLM (F2)

Fig. 7 details F2. Rendering the same eye as a color thickness map instead of a raw B-scan raises each stronger frontier VLM into the non-VLM reader band on discrimination and pulls it down on ECE and decision-flip together, while the small open/specialist VLMs do not recover. This is a cross-dataset comparison (the two axes use different datasets, vendors and label sources), so we read it as reader-specific, not as evidence that a thickness map is intrinsically a better input (§5).

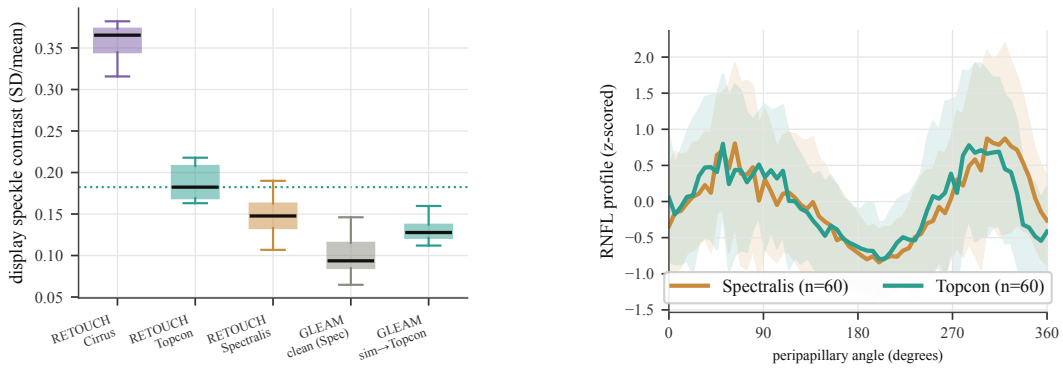


Figure 6: **Appearance/anatomy validity.** (a) Real per-vendor RETOUCH display speckle (Cirrus > Topcon > Spectralis) validates the frame-averaging direction and the calibrated sim moves GLEAM toward Topcon; (b) real cross-vendor cpRNFL: GLEAM Spectralis vs. GOALS Topcon profiles (single segmenter) agree in shape ($r \approx 0.93$; shape only, not device-interchangeability).

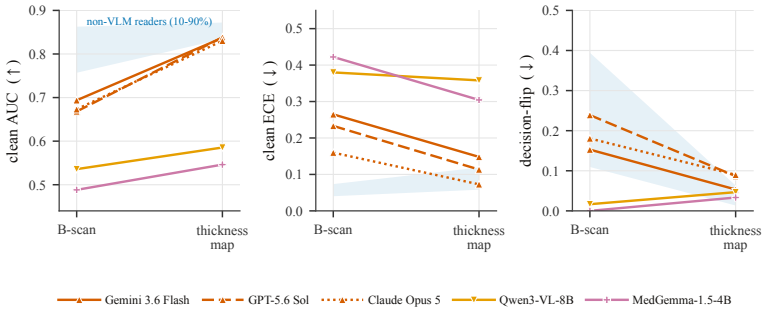


Figure 7: **The per-VLM thickness-map improvement, in three metrics.** Per-VLM change from the raw B-scan to the color thickness-map render (few-shot); the three frontier VLMs share the vermilion triangle and are distinguished by line style. The stronger frontier VLMs rise into the per-axis non-VLM reader band (10–90th percentile of the FM and CNN readers) on clean AUC and descend toward it on ECE and decision-flip together. The small open (Qwen3-VL-8B) and specialist (MedGemma-1.5-4B) models do not recover.

G Subgroup robustness

We ask whether the scanner-induced instability falls unevenly across disease severity or demographics, recomputing decision-flip per subgroup for every configuration on the seed-0 test split (`scripts/analyze_subgroups.py`). Fig. 8 aggregates this by the four capable reader classes (class mean and across-model spread; the near-chance open/specialist VLMs are near-constant and omitted from the figure), and Tables 4–5 give the per-reader point estimates for all configurations. Decision-flip stays broadly comparable across subgroups. No severity grade or demographic group shows a consistently elevated flip once the wide eye-level bootstrap CIs are accounted for. The one clear pattern is the B-scan over-referral of §4: for the appearance-keyed readers the *healthy* eyes flip most (MIRAGE · LP 0.62 vs. 0.28 moderate/advanced; RETFound · LP 0.47 vs. 0.09; Small-CNN 0.35 vs. 0.15), i.e. the false-referral load concentrates in the screening-relevant non-glaucoma population. Sensitivity change (Δsens) is likewise small and CI-overlapping across the glaucoma severity subgroups. *Caveats*: the GLEAM severity subgroups are small ($n \approx 57$–184 per grade, so 95% CIs span $\approx \pm 0.08$–0.13); **image quality is not recorded in either dataset** and GLEAM carries no demographics, so those subgroup axes cannot be tested and are left to future work.

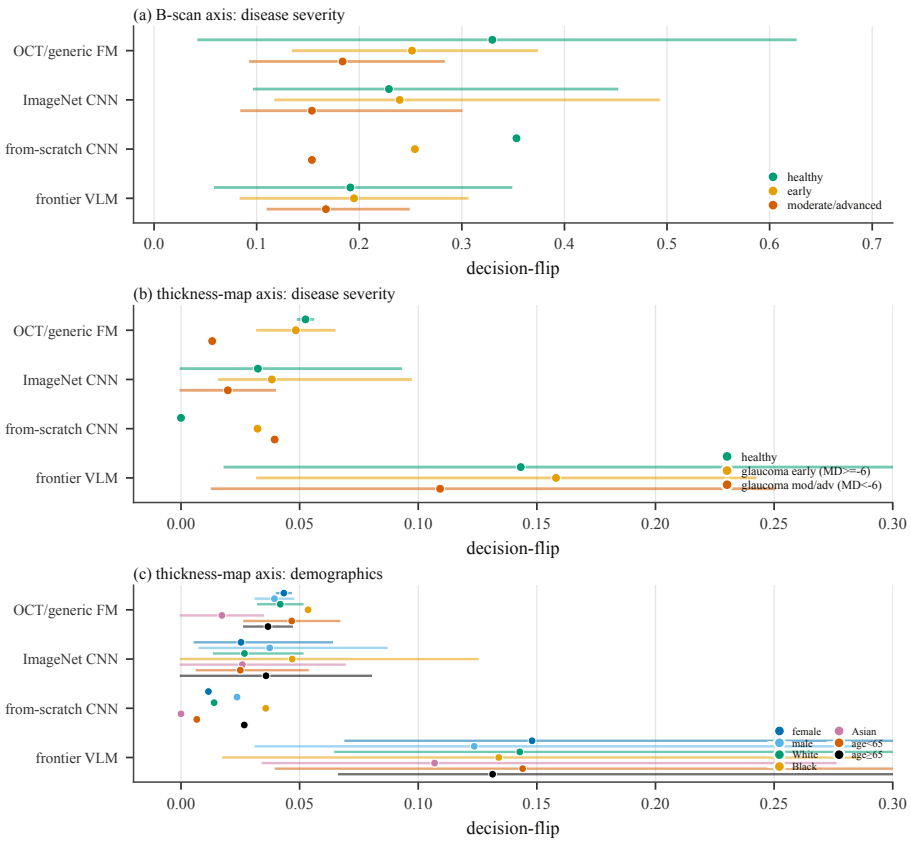


Figure 8: **Decision-flip by subgroup, aggregated by reader class** (the four capable Fig. 2 classes; the near-constant open/specialist VLMs are omitted here but listed in Tables 4–5). Markers are the class mean across its models; bars span the min–max across those models. **(a)** B-scan disease severity: for the appearance-keyed classes (OCT/generic FM, ImageNet CNN, from-scratch CNN) the healthy-eye flip is highest (over-referral on the screening population; the wide bars are within-class model spread, e.g. robust DINOv2 · FT vs. unstable MIRAGE · LP), whereas the frontier VLMs are flat. **(b)** Thickness-map severity and **(c)** map demographics: flip stays low and the subgroups cluster for every class, with no group consistently elevated. The flip scale differs between (a) and (b,c).

H Full per-axis tables

I Reproducibility and broader impact

Reproducibility. We release the full pipeline, the verbatim prompts (§B), the per-configuration model IDs and inference settings (Table 2), and all seeds. Splits are eye-level and seed-fixed; encoder probes are fit on the seed-0 split; VLM elicitation is greedy (single sample). Because frontier API models drift, we pin each model ID and API date and archive the raw per-eye model responses, so every table and figure reproduces exactly from the released code and responses; re-querying a live API may differ as the served model changes. On acceptance we release, under the MIT license, the exact synthetic-shift transforms (the B-scan appearance transform and the thickness-map scanner-offset, with every parameter and per-eye seed) together with the committed per-eye outputs, so a new model can be scored directly against the calibrated shift; the datasets remain under their own access terms (GLEAM; Harvard-GDP CC BY-NC-ND 4.0; GOALS; RETOUCH).

Broader impact. The deployment-relevant finding is a safety one: a reader can preserve AUC under a scanner change while its diagnostic decisions flip, so a naive AUC-only validation can license an unsafe glaucoma-diagnosis reader. We advocate decision-level, calibration-aware, shift-

Table 4: **Decision-flip by disease severity**, both axes, all configurations (point estimates; the per-reader eye-level bootstrap 95% CIs are wide and overlapping across grades, and the class-aggregated means and model spread are in Fig. 8a,b). Severity: B-scan from the GLEAM 4-class label (healthy / early / moderate-advanced); map from Harvard-GDP mean deviation (healthy / MD≥ -6 / MD<-6). "–" = reader not run on that axis.

	B-scan axis			Thickness-map axis		
Reader	healthy	early	mod/adv	healthy	early	mod/adv
DINOv2 · FT	0.04	0.14	0.13	0.05	0.03	0.01
DINOv2 · LP	0.18	0.14	0.24	0.06	0.06	0.01
MIRAGE · FT	0.51	0.34	0.25	–	–	–
MIRAGE · LP	0.62	0.37	0.28	–	–	–
RETFound · FT	0.15	0.17	0.11	–	–	–
RETFound · LP	0.47	0.36	0.09	–	–	–
ConvNeXt-B · FT	0.16	0.15	0.09	0.00	0.02	0.03
ConvNeXt-B · LP	0.10	0.15	0.09	0.02	0.02	0.00
ConvNeXt-L · FT	0.18	0.19	0.09	0.01	0.03	0.03
ConvNeXt-L · LP	0.17	0.19	0.15	0.04	0.06	0.01
ConvNeXt-T · FT	0.12	0.12	0.09	0.02	0.02	0.00
ConvNeXt-T · LP	0.41	0.37	0.16	0.01	0.03	0.03
ResNet-50 · FT	0.45	0.49	0.26	0.09	0.03	0.03
ResNet-50 · LP	0.23	0.25	0.30	0.06	0.10	0.04
SmallCNN (98k)	0.35	0.25	0.15	0.00	0.03	0.04
Claude Opus 5 *fs*	0.23	0.17	0.11	0.10	0.11	0.04
Claude Opus 5 *zs*	0.35	0.24	0.16	0.03	0.18	0.08
GPT-5.6 Sol *fs*	0.21	0.31	0.25	0.10	0.10	0.04
GPT-5.6 Sol *zs*	0.06	0.08	0.11	0.02	0.16	0.18
Gemini 2.5 Flash *fs*	0.20	0.29	0.19	0.30	0.23	0.09
Gemini 2.5 Flash *zs*	0.10	0.12	0.11	0.37	0.19	0.25
Gemini 2.5 Pro *fs*	0.27	0.19	0.21	0.17	0.18	0.09
Gemini 2.5 Pro *zs*	0.12	0.17	0.14	0.24	0.24	0.18
Gemini 3.6 Flash *fs*	0.14	0.20	0.15	0.08	0.03	0.01
Gemini 3.6 Flash *zs*	0.23	0.19	0.25	0.02	0.16	0.12
Qwen3-VL-8B *fs*	0.02	0.02	0.01	0.07	0.02	0.01
Qwen3-VL-8B *zs*	0.21	0.17	0.15	0.01	0.00	0.00
MedGemma-1.5-4B *fs*	0.00	0.00	0.00	0.02	0.08	0.01
MedGemma-1.5-4B *zs*	0.00	0.00	0.00	0.02	0.03	0.00

aware evaluation before deployment, and release the benchmark as a reusable check. The study uses de-identified public research datasets; no new patient data were collected.

Table 5: **Decision-flip by demographics** (thickness-map axis, Harvard-GDP; all configurations, point estimates, wide overlapping CIs; class-aggregated view in Fig. 8c). No group is consistently elevated. GLEAM (B-scan) carries no demographics.

	gender		race			age	
Reader	F	M	White	Black	Asian	<65	≥65
DINOv2 · FT	0.04	0.03	0.03	0.05	0.03	0.03	0.05
DINOv2 · LP	0.05	0.05	0.05	0.05	0.00	0.07	0.03
ConvNeXt-B · FT	0.01	0.01	0.01	0.00	0.00	0.02	0.00
ConvNeXt-B · LP	0.01	0.02	0.02	0.00	0.00	0.01	0.01
ConvNeXt-L · FT	0.01	0.02	0.01	0.04	0.00	0.01	0.03
ConvNeXt-L · LP	0.03	0.05	0.04	0.04	0.07	0.03	0.05
ConvNeXt-T · FT	0.01	0.02	0.01	0.04	0.00	0.01	0.02
ConvNeXt-T · LP	0.02	0.02	0.02	0.02	0.03	0.01	0.03
ResNet-50 · FT	0.06	0.06	0.05	0.12	0.07	0.05	0.08
ResNet-50 · LP	0.05	0.09	0.05	0.12	0.03	0.05	0.07
SmallCNN (98k)	0.01	0.02	0.01	0.04	0.00	0.01	0.03
Claude Opus 5 *fs*	0.11	0.06	0.08	0.11	0.10	0.09	0.09
Claude Opus 5 *zs*	0.08	0.07	0.07	0.11	0.03	0.08	0.07
GPT-5.6 Sol *fs*	0.11	0.06	0.08	0.11	0.07	0.10	0.07
GPT-5.6 Sol *zs*	0.09	0.09	0.08	0.14	0.03	0.09	0.09
Gemini 2.5 Flash *fs*	0.25	0.20	0.22	0.23	0.28	0.24	0.22
Gemini 2.5 Flash *zs*	0.32	0.28	0.33	0.29	0.17	0.30	0.31
Gemini 2.5 Pro *fs*	0.16	0.14	0.17	0.12	0.07	0.19	0.11
Gemini 2.5 Pro *zs*	0.24	0.21	0.26	0.14	0.17	0.25	0.20
Gemini 3.6 Flash *fs*	0.07	0.03	0.07	0.02	0.03	0.04	0.07
Gemini 3.6 Flash *zs*	0.07	0.08	0.07	0.07	0.10	0.07	0.08
Qwen3-VL-8B *fs*	0.06	0.03	0.05	0.05	0.03	0.07	0.03
Qwen3-VL-8B *zs*	0.00	0.01	0.00	0.00	0.00	0.01	0.00
MedGemma-1.5-4B *fs*	0.03	0.04	0.03	0.05	0.03	0.03	0.03
MedGemma-1.5-4B *zs*	0.03	0.01	0.01	0.00	0.10	0.03	0.01

Table 6: **Full B-scan (appearance) axis, all 29 configurations**. Spectralis → Topcon speckle shift (n=360 eyes). spec@shift = realized specificity under the shift at the 90%-specificity operating point (a low value signals decision collapse); ECE, aECE and Brier have no bootstrap CI in the released summaries. ECE is the positive-class calibration error (10 equal-width bins); aECE is the adaptive (equal-mass) ECE and Brier is the Brier score, a strictly proper score with no binning, so together they show the calibration reads are not a fixed-bin artifact. NCE = net calibration error, a signed calibration error [14] (mean predicted-class confidence minus accuracy; + = overconfident). All calibration reads are on the clean eyes.

Reader	Class	AUC [95%, CI]	Flip [95%, CI]	Δsens [95%, CI]	spec@shift	ECE	aECE	Brier	NCE	ΔAUC
DINOv2 · FT	FM	0.876 [0.84, 0.91]	0.086 [0.06, 0.12]	+0.080 [0.03, 0.13]	0.92	0.052	0.073	0.145	-0.003	+0.011
RETFound · FT	FM	0.847 [0.80, 0.89]	0.142 [0.11, 0.18]	-0.091 [-0.14, -0.04]	0.81	0.072	0.080	0.158	+0.067	-0.007
RETFound · LP	FM	0.845 [0.80, 0.88]	0.328 [0.28, 0.38]	-0.227 [-0.29, -0.16]	0.52	0.040	0.035	0.160	-0.001	+0.061
DINOv2 · LP	FM	0.842 [0.80, 0.88]	0.194 [0.16, 0.24]	+0.165 [0.10, 0.23]	0.92	0.047	0.039	0.162	+0.034	+0.059
MIRAGE · FT	FM	0.736 [0.68, 0.78]	0.394 [0.34, 0.44]	-0.386 [-0.46, -0.32]	0.61	0.121	0.119	0.220	+0.097	-0.002
MIRAGE · LP	FM	0.701 [0.65, 0.76]	0.472 [0.42, 0.52]	-0.722 [-0.79, -0.65]	0.07	0.041	0.067	0.221	-0.021	+0.044
ConvNeXt-T · FT	ImageNet CNN	0.869 [0.83, 0.90]	0.111 [0.08, 0.14]	-0.097 [-0.14, -0.06]	0.68	0.067	0.070	0.147	-0.003	+0.026
ConvNeXt-B · FT	ImageNet CNN	0.854 [0.81, 0.89]	0.136 [0.10, 0.17]	-0.051 [-0.10, 0.01]	0.74	0.049	0.067	0.155	-0.019	+0.050
ConvNeXt-L · FT	ImageNet CNN	0.836 [0.79, 0.87]	0.153 [0.12, 0.19]	-0.068 [-0.12, -0.02]	0.76	0.066	0.067	0.164	+0.051	+0.045
ResNet-50 · FT	ImageNet CNN	0.833 [0.78, 0.87]	0.394 [0.34, 0.44]	-0.307 [-0.37, -0.24]	0.55	0.068	0.078	0.169	+0.035	+0.006
ConvNeXt-B · LP	ImageNet CNN	0.826 [0.78, 0.87]	0.108 [0.07, 0.14]	+0.068 [0.02, 0.13]	0.91	0.070	0.055	0.170	+0.029	+0.038
ConvNeXt-T · LP	ImageNet CNN	0.826 [0.78, 0.87]	0.325 [0.28, 0.38]	-0.199 [-0.26, -0.14]	0.60	0.036	0.051	0.169	+0.031	+0.038
ConvNeXt-L · LP	ImageNet CNN	0.825 [0.78, 0.86]	0.167 [0.13, 0.21]	+0.028 [-0.03, 0.09]	0.83	0.062	0.058	0.168	+0.014	+0.048
ResNet-50 · LP	ImageNet CNN	0.825 [0.78, 0.86]	0.258 [0.22, 0.30]	+0.136 [0.06, 0.21]	0.89	0.075	0.076	0.174	+0.027	+0.080
SmallCNN (98k)	scratch CNN	0.788 [0.74, 0.83]	0.272 [0.23, 0.32]	-0.324 [-0.40, -0.25]	0.65	0.042	0.063	0.189	-0.039	+0.017
Gemini 3.6 Flash *fs*	frontier VLM	0.694 [0.64, 0.74]	0.153 [0.12, 0.19]	+0.006 [-0.03, 0.04]	0.90	0.265	0.260	0.292	+0.258	-0.001
Claude Opus 5 *fs*	frontier VLM	0.673 [0.62, 0.73]	0.181 [0.14, 0.22]	+0.045 [0.00, 0.09]	0.97	0.160	0.159	0.253	+0.132	+0.030
GPT-5.6 Sol *fs*	frontier VLM	0.668 [0.61, 0.73]	0.239 [0.19, 0.28]	+0.108 [0.05, 0.17]	0.92	0.233	0.218	0.283	+0.228	-0.012
Gemini 2.5 Pro *fs*	frontier VLM	0.635 [0.58, 0.69]	0.239 [0.20, 0.28]	-0.017 [-0.06, 0.02]	0.99	0.324	0.313	0.338	+0.322	-0.004
GPT-5.6 Sol *zs*	frontier VLM	0.631 [0.58, 0.69]	0.081 [0.06, 0.11]	+0.045 [-0.02, 0.11]	0.92	0.307	0.297	0.337	+0.304	-0.004
Gemini 2.5 Flash *fs*	frontier VLM	0.622 [0.57, 0.68]	0.211 [0.17, 0.25]	+0.040 [-0.01, 0.09]	0.95	0.297	0.279	0.325	+0.295	+0.013
Claude Opus 5 *zs*	frontier VLM	0.607 [0.55, 0.66]	0.269 [0.22, 0.31]	-0.034 [-0.09, 0.02]	0.97	0.113	0.090	0.250	+0.076	+0.010
Gemini 3.6 Flash *zs*	frontier VLM	0.596 [0.54, 0.65]	0.228 [0.19, 0.27]	-0.011 [-0.08, 0.05]	0.89	0.356	0.308	0.368	+0.349	+0.032
Gemini 2.5 Flash *zs*	frontier VLM	0.571 [0.52, 0.62]	0.108 [0.08, 0.14]	+0.023 [-0.03, 0.08]	0.91	0.390	0.372	0.399	+0.379	+0.053
Gemini 2.5 Pro *zs*	frontier VLM	0.515 [0.47, 0.56]	0.133 [0.10, 0.17]	+0.023 [-0.04, 0.09]	0.92	0.422	0.407	0.427	+0.420	-0.029
Qwen3-VL-8B *fs*	open VLM	0.536 [0.49, 0.58]	0.017 [0.01, 0.03]	+0.000 [0.00, 0.00]	1.00	0.380	0.369	0.393	+0.380	+0.000
Qwen3-VL-8B *zs*	open VLM	0.462 [0.42, 0.50]	0.186 [0.15, 0.23]	+0.000 [0.00, 0.00]	1.00	0.510	0.487	0.510	+0.510	+0.038
MedGemma-1.5-4B *zs*	specialist VLM	0.539 [0.49, 0.58]	0.000 [0.00, 0.00]	+0.023 [-0.01, 0.05]	0.99	0.464	0.465	0.463	+0.465	-0.015
MedGemma-1.5-4B *fs*	specialist VLM	0.488 [0.43, 0.54]	0.000 [0.00, 0.00]	+0.034 [0.00, 0.07]	0.98	0.422	0.422	0.430	+0.422	-0.036

Table 7: **Full thickness-map (color-map) axis, all 25 configurations**. Cirrus → Spectralis offset; OCT-specific B-scan FMs are not run on map renders (n=300 eyes). spec@shift = realized specificity under the shift at the 90%-specificity operating point (a low value signals decision collapse); ECE, aECE and Brier have no bootstrap CI in the released summaries. ECE is the positive-class calibration error (10 equal-width bins); aECE is the adaptive (equal-mass) ECE and Brier is the Brier score, a strictly proper score with no binning, so together they show the calibration reads are not a fixed-bin artifact. NCE = net calibration error, a signed calibration error [14] (mean predicted-class confidence minus accuracy; + = overconfident). All calibration reads are on the clean eyes.

Reader	Class	AUC [95%, CI]	Flip [95%, CI]	Δsens [95%, CI]	spec@shift	ECE	aECE	Brier	NCE	ΔAUC
DINOv2 · FT	FM	0.876 [0.83, 0.91]	0.037 [0.02, 0.06]	+0.014 [-0.01, 0.04]	0.93	0.090	0.089	0.136	-0.018	+0.004
DINOv2 · LP	FM	0.849 [0.80, 0.89]	0.047 [0.02, 0.07]	+0.029 [-0.01, 0.07]	0.90	0.078	0.062	0.153	+0.035	-0.001
ConvNeXt-L · FT	ImageNet CNN	0.872 [0.83, 0.91]	0.017 [0.00, 0.03]	+0.007 [-0.02, 0.03]	0.91	0.067	0.074	0.137	-0.040	+0.001
ResNet-50 · FT	ImageNet CNN	0.865 [0.82, 0.91]	0.063 [0.04, 0.09]	+0.007 [-0.02, 0.03]	0.96	0.135	0.130	0.152	-0.085	-0.005
ConvNeXt-T · FT	ImageNet CNN	0.862 [0.82, 0.91]	0.017 [0.00, 0.03]	+0.000 [-0.03, 0.03]	0.91	0.095	0.098	0.141	-0.037	-0.000
ConvNeXt-B · FT	ImageNet CNN	0.859 [0.81, 0.90]	0.010 [0.00, 0.02]	+0.000 [0.00, 0.00]	0.91	0.112	0.169	0.170	+0.011	-0.001
ConvNeXt-B · LP	ImageNet CNN	0.843 [0.79, 0.89]	0.013 [0.00, 0.03]	-0.014 [-0.04, 0.01]	0.90	0.057	0.065	0.151	+0.021	-0.001
ConvNeXt-T · LP	ImageNet CNN	0.836 [0.79, 0.88]	0.020 [0.01, 0.04]	+0.058 [0.02, 0.10]	0.93	0.058	0.066	0.155	+0.001	-0.001
ConvNeXt-L · LP	ImageNet CNN	0.834 [0.78, 0.88]	0.040 [0.02, 0.06]	-0.007 [-0.03, 0.02]	0.89	0.081	0.054	0.156	+0.059	-0.008
ResNet-50 · LP	ImageNet CNN	0.831 [0.78, 0.88]	0.063 [0.04, 0.09]	-0.007 [-0.04, 0.03]	0.88	0.058	0.067	0.163	+0.029	+0.014
SmallCNN (98k)	scratch CNN	0.847 [0.80, 0.89]	0.017 [0.00, 0.03]	+0.000 [-0.04, 0.04]	0.94	0.119	0.139	0.169	+0.033	-0.006
Claude Opus 5 *zs*	frontier VLM	0.850 [0.81, 0.89]	0.073 [0.05, 0.10]	+0.058 [0.00, 0.12]	0.93	0.106	0.113	0.167	-0.011	+0.023
GPT-5.6 Sol *fs*	frontier VLM	0.838 [0.79, 0.88]	0.087 [0.06, 0.12]	+0.022 [-0.03, 0.07]	0.93	0.113	0.110	0.170	+0.110	-0.005
Gemini 3.6 Flash *fs*	frontier VLM	0.837 [0.79, 0.88]	0.053 [0.03, 0.08]	+0.058 [0.02, 0.09]	0.93	0.148	0.140	0.175	+0.134	+0.016
Claude Opus 5 *fs*	frontier VLM	0.830 [0.78, 0.88]	0.090 [0.06, 0.12]	+0.029 [-0.01, 0.07]	0.93	0.073	0.074	0.161	+0.014	-0.002
Gemini 2.5 Pro *fs*	frontier VLM	0.807 [0.76, 0.86]	0.150 [0.11, 0.19]	-0.051 [-0.12, 0.01]	0.96	0.150	0.153	0.189	+0.150	-0.007
Gemini 3.6 Flash *zs*	frontier VLM	0.797 [0.75, 0.85]	0.073 [0.05, 0.10]	+0.043 [-0.01, 0.11]	0.94	0.230	0.226	0.239	+0.201	-0.006
Gemini 2.5 Flash *fs*	frontier VLM	0.775 [0.72, 0.82]	0.230 [0.19, 0.28]	-0.029 [-0.11, 0.04]	0.94	0.173	0.147	0.218	+0.169	-0.001
GPT-5.6 Sol *zs*	frontier VLM	0.772 [0.72, 0.82]	0.090 [0.06, 0.12]	-0.014 [-0.09, 0.06]	0.90	0.233	0.227	0.241	+0.170	-0.026
Gemini 2.5 Pro *zs*	frontier VLM	0.713 [0.66, 0.77]	0.227 [0.18, 0.28]	-0.007 [-0.09, 0.08]	0.98	0.268	0.261	0.282	+0.266	+0.004
Gemini 2.5 Flash *zs*	frontier VLM	0.644 [0.58, 0.71]	0.303 [0.26, 0.35]	+0.036 [-0.03, 0.10]	1.00	0.229	0.221	0.287	+0.211	-0.026
Qwen3-VL-8B *fs*	open VLM	0.585 [0.55, 0.62]	0.047 [0.02, 0.07]	+0.000 [-0.04, 0.04]	0.94	0.358	0.345	0.369	+0.358	+0.018
Qwen3-VL-8B *zs*	open VLM	0.500 [0.50, 0.50]	0.003 [0.00, 0.01]	+0.000 [0.00, 0.00]	1.00	0.490	0.490	0.489	+0.490	-0.003
MedGemma-1.5-4B *fs*	specialist VLM	0.546 [0.48, 0.61]	0.033 [0.02, 0.06]	+0.000 [-0.04, 0.04]	0.92	0.304	0.291	0.341	+0.299	+0.025
MedGemma-1.5-4B *zs*	specialist VLM	0.411 [0.35, 0.47]	0.020 [0.01, 0.04]	-0.014 [-0.04, 0.00]	0.95	0.384	0.364	0.406	+0.379	+0.018

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